The Matthews Resonance Framework

A Deterministic Nonlinear Phase-Ratio Stabilization System
with Self-Correcting Triune Harmonic Architecture

Patent Pending.

Abstract

We present a deterministic nonlinear dynamical framework for rational phase-ratio stabilization in coupled oscillatory systems. A reduced residual phase equation yields an explicit nonlinear capture boundary for pairwise rational locking. We then construct a dynamically closed triune harmonic architecture by introducing triadic phase feedback via an integer-resonance residual. This produces a second capture surface governing triadic coherence. We analyze equilibrium existence, local stability, and coupled residual dynamics, and provide a composite Lyapunov structure certifying monotone dissipation of the triadic residual under its feedback potential. We characterize parameter regimes for pair-only, triune-only, and dual locking, and give an explicit falsification protocol. Numerical correspondence tests are specified to verify the predicted capture boundaries and regime transitions.

1 Scope and Notation

We work on phases $\phi_i(t) \in \mathbb{R}$ with physical phase identified modulo 2π, i.e. on the torus $\mathbb{T}^n$. The model is *deterministic*: all dynamics are governed by autonomous ordinary differential equations (ODEs) with no stochastic forcing.

Let $p, q \in \mathbb{Z}_{>0}$ be coprime integers defining a target rational phase relationship between two oscillators. Define the *pairwise rational residual*

$$\Delta \;:=\; q\phi_2 - p\phi_1. \tag{1}$$

Define the *triadic residual* associated with an integer resonance vector $\boldsymbol{k} \in \mathbb{Z}^3$. In the minimal triune ladder used here we take

$$\boldsymbol{k} = (3, -1, -1), \qquad \Psi \;:=\; 3\phi_0 - \phi_1 - \phi_2. \tag{2}$$

Throughout, $\epsilon > 0$ denotes pairwise coupling strength and $\kappa > 0$ denotes triadic feedback strength.

2 Two-Phase Nonlinear Rational Phase Locking

2.1 Model

Consider the two-phase coupled system

$$\dot{\phi}_1 = \omega_1 + \epsilon\, q\, \sin(\Delta), \tag{3}$$

$$\dot{\phi}_2 = \omega_2 - \epsilon\, p\, \sin(\Delta), \tag{4}$$

with $\Delta = q\phi_2 - p\phi_1$.

2.2 Reduced residual dynamics

Differentiating Δ yields the reduced scalar ODE

$$\dot{\Delta} = D_{\text{pair}} - \epsilon(p^2 + q^2)\sin(\Delta), \qquad D_{\text{pair}} := q\omega_2 - p\omega_1. \tag{5}$$

2.3 Capture boundary and equilibrium existence

Theorem 1 (Nonlinear pair capture boundary)**.** *Equation* (5) *admits an equilibrium* Δ^* *if and only if*

$$|D_{pair}| \leq \epsilon(p^2 + q^2). \tag{6}$$

When (6) *holds with strict inequality, equilibria satisfy* $\sin(\Delta^*) = D_{pair}/(\epsilon(p^2 + q^2))$.

Proof. An equilibrium requires $D_{\text{pair}} = \epsilon(p^2+q^2)\sin(\Delta^*)$. Since $|\sin(\cdot)| \leq 1$, existence is equivalent to (6). □

2.4 Local stability

Proposition 1 (Local asymptotic stability of pair equilibrium)**.** *Assume* (6) *holds strictly so that* Δ^* *exists with* $|\sin(\Delta^*)| < 1$. *Then any equilibrium with* $\cos(\Delta^*) > 0$ *is locally asymptotically stable for* $\epsilon > 0$.

Proof. Linearize (5) about Δ^*:

$$\dot{\delta} = -\epsilon(p^2 + q^2)\cos(\Delta^*)\,\delta,$$

so $\delta(t) \to 0$ exponentially iff $\cos(\Delta^*) > 0$. □

Remark 1. *The inequality* (6) *is a classical Arnold-tongue capture condition: outside the boundary,* Δ *drifts on* $\mathbb{T}$ *and no stationary phase residual exists.*

3 Triune Harmonic Architecture and Dynamic Closure

3.1 Triune ladder (frequency structure)

Introduce a base oscillator ϕ_0 with frequency ω_0. A canonical triune ladder is

$$\omega_1 = 2\omega_0, \qquad \omega_2 = 3\omega_0, \tag{7}$$

which implies the integer resonance

$$3\omega_0 - \omega_1 - \omega_2 = 0. \tag{8}$$

3.2 Triadic residual and feedback potential

Define the triadic residual

$$\Psi = 3\phi_0 - \phi_1 - \phi_2. \tag{9}$$

To make the triune ladder *self-correcting*, we introduce a feedback potential

$$V(\Psi) = -\kappa\cos(\Psi), \qquad \kappa > 0, \tag{10}$$

and apply its gradient to the phase flows in proportions matching $\boldsymbol{k} = (3, -1, -1)$.

3.3 Full self-correcting triune system

The complete closed system is

$$\dot{\phi}_0 = \omega_0 + 3\kappa \sin(\Psi), \tag{11}$$
$$\dot{\phi}_1 = \omega_1 + \epsilon\, q\, \sin(\Delta) - \kappa \sin(\Psi), \tag{12}$$
$$\dot{\phi}_2 = \omega_2 - \epsilon\, p\, \sin(\Delta) - \kappa \sin(\Psi), \tag{13}$$

with

$$\Delta = q\phi_2 - p\phi_1, \qquad \Psi = 3\phi_0 - \phi_1 - \phi_2. \tag{14}$$

This system is autonomous, smooth, and deterministic on $\mathbb{T}^3$.

4 Triune Reduced Dynamics and Capture Boundary

Differentiate Ψ using (11)–(13):

$$\begin{aligned}\dot{\Psi} &= 3\dot{\phi}_0 - \dot{\phi}_1 - \dot{\phi}_2 \\ &= \big(3\omega_0 - \omega_1 - \omega_2\big) - \epsilon\,(q-p)\sin(\Delta) - 7\kappa \sin(\Psi).\end{aligned} \tag{15}$$

Define triadic detuning

$$D_{\text{triune}} \coloneqq 3\omega_0 - \omega_1 - \omega_2. \tag{16}$$

4.1 Decoupled triune capture (balanced case)

If the triune architecture is used with $p = q$ (balanced rational residual), then the cross-term in (15) vanishes and Ψ obeys a closed scalar equation:

$$\dot{\Psi} = D_{\text{triune}} - 7\kappa \sin(\Psi). \tag{17}$$

This yields an explicit triune capture boundary.

Theorem 2 (Triune capture boundary (balanced case))**.** *Assume $p = q$ so that (17) holds. Then Ψ admits an equilibrium Ψ^* if and only if*

$$|D_{triune}| \le 7\kappa. \tag{18}$$

Proof. Same argument as Theorem 1 applied to (17). □

subsectionCoupled triune capture (general p, q) For general $p \neq q$, the triune residual is coupled to Δ through the term $-\epsilon(q-p)\sin(\Delta)$ in (15). Since $\sin(\Delta) \in [-1, 1]$, the coupling contributes a bounded forcing of magnitude $\epsilon|q-p|$. Thus, a sufficient capture condition is obtained by requiring that the triadic restoring term can dominate worst-case forcing:

Proposition 2 (Sufficient triune capture under bounded coupling)**.** *If*

$$|D_{triune}| + \epsilon|q-p| \le 7\kappa, \tag{19}$$

then there exists at least one phase Ψ^* for which $\dot{\Psi} = 0$ is attainable for some Δ and the triadic residual admits bounded trapping (i.e. the triadic restoring term can counteract both detuning and maximal pair-coupling forcing).

Proof. From (15),

$$\dot{\Psi} = D_{\text{triune}} - \epsilon(q-p)\sin(\Delta) - 7\kappa\sin(\Psi).$$

The first two terms together lie in the interval $[D_{\text{triune}} - \epsilon|q-p|,\ D_{\text{triune}} + \epsilon|q-p|]$. If (19) holds then this interval lies within $[-7\kappa, 7\kappa]$. Hence there exists Ψ^* with $\sin(\Psi^*)$ matching the required value for some admissible $\sin(\Delta) \in [-1, 1]$. □

Remark 2. *Condition* (19) *is conservative. In many trajectories* $\sin(\Delta)$ *is not adversarial and the effective forcing is smaller, enlarging the observed triune capture region.*

5 Dual Capture Architecture and Regime Map

The full system exhibits distinct regimes depending on whether the pair residual Δ and the triadic residual Ψ admit equilibria/trapping.

5.1 Pair capture surface

Pair capture is governed by Theorem 1:

$$|D_{\text{pair}}| \leq \epsilon(p^2 + q^2).$$

5.2 Triune capture surface

Triune capture is governed by:

- exact condition (18) in the balanced decoupled case ($p = q$),
- sufficient condition (19) in the general coupled case.

5.3 Regimes

Define:

Dual-lock regime
Pair capture holds and triune capture holds (exact or sufficient). Both Δ and Ψ can stabilize.

Pair-only regime
Pair capture holds but triune capture fails. Rational locking occurs without triadic coherence.

Triune-only regime
Triune capture holds but pair capture fails. Triadic coherence can persist even if rational pair locking is lost (depending on coupling).

Drift regime
Both capture conditions fail. Both residuals drift on $\mathbb{T}$.

6 Lyapunov Structure for Triadic Self-Correction

A central purpose of the triadic feedback (10) is to impose monotone dissipation in the triadic residual.

6.1 Triadic Lyapunov function (balanced case)

Assume $p = q$ so that Ψ obeys (17). Define the shifted potential

$$\mathcal{L}_\Psi(\Psi) := 7\kappa\big(1 - \cos\Psi\big) - D_{\text{triune}}\Psi. \tag{20}$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{L}_\Psi(\Psi(t)) = \big(7\kappa\sin\Psi - D_{\text{triune}}\big)\dot{\Psi} = -\dot{\Psi}^{\,2} \le 0. \tag{21}$$

Theorem 3 (Monotone dissipation of triadic residual (balanced case))**.** *If $p = q$, then $\mathcal{L}_\Psi$ is a Lyapunov function for the triadic reduced dynamics (17); along any trajectory $\mathcal{L}_\Psi$ is non-increasing and strictly decreasing except at equilibria. In particular, if (18) holds strictly, $\Psi(t)$ converges to the stable equilibrium set.*

Proof. Equation (21) follows by direct differentiation and substitution of (17). □

Remark 3. *In the general coupled case ($p \neq q$), Ψ experiences bounded forcing from Δ (see (15)). Then $\mathcal{L}_\Psi$ becomes an input-to-state Lyapunov candidate and yields boundedness/trapping statements under sufficient restoring strength, consistent with Proposition 2.*

7 Simulation Correspondence and Verification Protocol

The framework is designed to be tested by direct numerical integration of (11)–(13) and/or by reduced residual dynamics (5), (17) (balanced case) and (15) (general case).

7.1 Recommended observables

Define wrapped residuals on $[-\pi, \pi]$:

$$\mathrm{wrap}(x) = (x + \pi) \bmod (2\pi) - \pi, \quad \Delta_w = \mathrm{wrap}(\Delta), \quad \Psi_w = \mathrm{wrap}(\Psi).$$

Define phase-locking value (PLV) over a window $[t - T, t]$:

$$\mathrm{PLV}_X(t; T) := \left|\frac{1}{T}\int_{t-T}^{t} e^{iX(s)}\,\mathrm{d}s\right|, \qquad X \in \{\Delta, \Psi\}. \tag{22}$$

In discrete simulation with step δt, use the empirical mean over the window.

7.2 Boundary test (pair)

Fix p, q, ω_1, ω_2 and compute

$$D_{\text{pair}} = q\omega_2 - p\omega_1, \qquad \epsilon_{\text{crit}} = \frac{|D_{\text{pair}}|}{p^2 + q^2}.$$

Run two simulations:

$$\epsilon = 0.9\,\epsilon_{\text{crit}} \quad \text{and} \quad \epsilon = 1.1\,\epsilon_{\text{crit}}.$$

Prediction:

- below critical: Δ_w drifts and PLV_Δ is low,
- above critical: Δ_w stabilizes to a constant and $\mathrm{PLV}_\Delta \approx 1$.

7.3 Boundary test (triune)

Balanced case ($p = q$): compute

$$D_{\text{triune}} = 3\omega_0 - \omega_1 - \omega_2, \qquad \kappa_{\text{crit}} = \frac{|D_{\text{triune}}|}{7}.$$

Run $\kappa = 0.9\,\kappa_{\text{crit}}$ and $\kappa = 1.1\,\kappa_{\text{crit}}$ and test Ψ_w, PLV_Ψ.

General case ($p \neq q$): use the sufficient bound (19) as a conservative target.

7.4 Detuning step test (unlock / recapture)

Apply a piecewise detuning schedule (e.g. drift $\omega_1(t)$ or $\omega_2(t)$) so that the trajectory crosses the predicted capture boundary:

- inside capture region: PLV_Δ high,
- outside capture region: PLV_Δ drops,
- upon return: PLV_Δ recovers (recapture).

This is expected and is not a contradiction; it is the operational meaning of the capture surface.

7.5 Figures (optional)

If you have generated plots, place the image files in your Overleaf project and uncomment the figure blocks below.

Figure 1: Pair residual Δ (wrapped) under a detuning step: comparison of paper-style pair-only behavior vs. triune-feedback system.

Figure 2: Triadic residual Ψ (wrapped) under a detuning step: triune-feedback stabilizes Ψ while a non-feedback model drifts.

Figure 3: PLV metrics PLV_Δ and PLV_Ψ (sliding window). Dual-lock appears as simultaneous high PLV for both residuals.

Figure 4: Arnold-tongue slice: PLV_Δ vs. ϵ for fixed (p, q) and detuning, showing a sharp transition near $\epsilon_{\text{crit}} = |D_{\text{pair}}|/(p^2 + q^2)$.

8 Falsification Protocol

The framework is falsified if any of the following occur under controlled numerical integration (with sufficiently small step/tolerance to rule out solver artifacts):

1. **Pair false-positive:** sustained pair locking (high PLV_Δ and bounded Δ_w) occurs when $|D_{\mathrm{pair}}| > \epsilon(p^2+q^2)$.
2. **Pair false-negative:** pair locking fails (low PLV_Δ and drifting Δ_w) when $|D_{\mathrm{pair}}| \le \epsilon(p^2+q^2)$.
3. **Triune false-positive (balanced case):** sustained triune locking occurs when $|D_{\mathrm{triune}}| > 7\kappa$.
4. **Triune false-negative (balanced case):** triune locking fails when $|D_{\mathrm{triune}}| \le 7\kappa$.
5. **Coupled inconsistency:** in the general coupled case, observed triune trapping contradicts the bounded-forcing structure of (15) across repeated solver/timestep settings.

9 Conclusion

We have defined a deterministic nonlinear phase-ratio stabilization system with: (i) a rational residual Δ producing an explicit pairwise capture boundary, and (ii) a triadic residual Ψ closed by a feedback potential producing a second capture surface and self-correction of the triune harmonic ladder. The resulting dual-capture architecture partitions parameter space into regimes of dual-lock, pair-only, triune-only, and drift. A Lyapunov function certifies monotone dissipation of the triadic residual in the balanced case, and bounded-forcing analysis provides conservative capture guarantees in the general coupled case. The framework is directly testable by the falsification protocol herein.

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A Appendix A: Minimal implementation notes (for reproducibility)

ODE integration

Integrate (11)–(13) with an adaptive solver (e.g. RK45, Radau) and strict tolerances (e.g. `rtol` $= 10^{-9}$, `atol` $= 10^{-12}$) to suppress numerical unlocking artifacts.

Runtime selection

There is no intrinsic "runtime" in the ODE. A practical runtime is determined by the linearized convergence rates. Near a stable pair equilibrium, the local decay rate is on the order of $\epsilon(p^2+q^2)$ (up to $\cos(\Delta^*)$ factors); near a stable triune equilibrium (balanced case) the decay rate is on the order of 7κ (up to $\cos(\Psi^*)$). A robust choice is to simulate for $t_{\mathrm{end}} \gtrsim 10/\min\{\epsilon(p^2+q^2),\, 7\kappa\}$ in regimes expected to lock, and longer in step-detuning tests to observe unlock/recapture transitions.

Lock decision

A conservative empirical lock decision on a terminal interval can be defined by:

$$\mathrm{PLV}_\Delta > 0.95 \text{ and } \mathrm{std}(\Delta_w) < 0.25 \text{ rad}$$

(and similarly for Ψ), computed over the last 20% of the run.